

Evolutionary features of new picorna-like viruses in *Culex* (*Melanoconion*) mosquitoes

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ABSTRACT

In this study, we investigated unclassified picorna-like viruses in *Culex* (*Melanoconion*) mosquitoes from São Paulo, Brazil, an area of high mosquito biodiversity and arbovirus activity. Two mosquito pools were processed using next-generation sequencing (NGS), and datasets were analyzed *via de novo* assembly to reconstruct viral genomes and assess evolutionary relationships. We identified two highly similar viral genomes, named *Culex* (*Melanoconion*) picorna-like virus, CmV_B38 and CmV_B39, exhibiting 99.93% nucleotide identity, both of which encode a three-domain replication block characteristic of viruses within the order *Picornavirales*. Phylogenetic reconstruction based on the RNA-dependent RNA polymerase (RdRp) gene revealed that these viruses form a distinct, previously undescribed clade, most closely related to Yongsan picorna-like virus 4 and several other unclassified viruses that have been reported predominantly in Asian regions. These findings may indicate possible geographical connectivity or convergence in viral evolution across distinct ecosystems. Notably, the results highlight the underexplored diversity of insect-specific viruses, particularly those associated with mosquito vectors. Furthermore, the data are consistent with the hypothesis that ecological factors and host specificity could influence the evolutionary dynamics of these viral lineages. The study not only enhances our understanding of the mosquito-associated virome but also emphasizes the critical need for ongoing viral surveillance, especially in biodiverse regions. Such efforts are essential for elucidating the evolutionary dynamics of RNA viruses and for anticipating the emergence of novel viral pathogens that may pose future risks to public health or agriculture.

1. Introduction

Viruses of the order *Picornavirales* include viruses that infect a wide range of hosts, including mosquitoes (Le Gall et al., 2008; Cholleti et al., 2018). According to the most recent classification of the International Committee on Taxonomy of Viruses (ICTV), the order comprises nine recognized families: *Caliciviridae*, *Dicistroviridae*, *Iflaviridae*, *Marnaviridae*, *Noraviridae*, *Picornaviridae*, *Polycipiviridae*, *Secoviridae*, and *Solinviridae* (Le Gall et al., 2008).

The genomes of members of the order *Picornavirales* range from 7.2 to 9.8 kb, sharing common features such as a single-stranded positive-sense RNA genome that contains the three-domain replication block (Helicase (Hel), protease (Pro), and RNA-dependent RNA polymerase (RdRP)) (Le Gall et al., 2008), which is highly conserved and useful for viral taxonomy (Poch et al., 1989). The polyprotein also encodes structural proteins (VPs) that form the viral capsid (Le Gall et al., 2008).

In recent years, the viral metagenomics revolution has revealed a diversity of *Picornavirales*-like viruses, not yet officially classified.

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Notable among these are picorna-like viruses (PLVs) and posa-like viruses. These viruses have been found in a wide variety of hosts, and many of these viruses show less than 40% amino-acid identity with recognized families, making classification difficult and impossible to assign new viruses to existing families (Shi et al., 2016; Oude Munnink et al., 2017; Cholleti et al., 2018; Zell et al., 2022, 2024).

Examples of new *Picornavirales* include posavirus initially identified in pig feces (Chen et al., 2018; Aoki et al., 2019), husavirus (human feces) (Oude Munnink et al., 2015; Han et al., 2020; Ramos et al., 2021; Demoliner et al., 2023), pansavirus (panda feces) (Zhang et al., 2017), basavirus (bat feces) (Waller et al., 2024), rasavirus (rat feces) (Oude Munnink et al., 2017) and fisavirus (fish) (Reuter et al., 2015).

Furthermore, new provisional groupings, such as aquatic picorna-like viruses and insect-specific picorna-like viruses, have been proposed, reflecting distinct ecological patterns of these viruses (Bolling et al., 2015; Shi et al., 2016). To accommodate this diversity, molecular phylogeny and genetic distance criteria have been suggested to define new taxa (Simmonds et al., 2023).

Within this expanding landscape of viral diversity, mosquitoes of the family Culicidae stand out as well-known vectors of arboviruses and hosts of a wide range of insect-specific and non-insect-specific viruses worldwide. The family comprises 3729 extant species, classified into two subfamilies: Anophelinae, with three genera, and Culicinae, with 110 genera divided into 11 tribes (Da Silva et al., 2020). The genus *Culex* is recognized not only for its role in arbovirus transmission but also as a natural host of several insect-specific viruses with no known involvement with vertebrates (Bolling et al., 2015).

Recent studies showed that these mosquitoes harbor a wide variety of viruses, including those belonging to the order *Picornavirales* (Shi et al., 2016; Cholleti et al., 2018). Within the genus *Culex*, the subgenus *Melanoconion* is subdivided into two sections: *Melanoconion* and *Spisses* (Sá et al., 2020). This subgenus is particularly relevant for including species that can act as vectors for viruses such as Venezuelan equine encephalitis and West Nile virus (Turell et al., 2006; Cupp et al., 2007; Barrera et al., 2011). Our previous study conducted viral metagenomic analysis from adult female mosquitoes collected at the São Paulo Zoo Foundation, Brazil, aiming to contribute to the understanding of mosquito-associated viral diversity (Guimarães et al., 2025). In light of this, this study is specifically a more in-depth exploration of new picorna-like viruses identified in *Culex* (*Melanoconion*) mosquitoes.

2. Materials and methods

2.1. Sample collection and mosquito identification

Two pools (pool B38 and B39), each containing 50 adult non-engorged female specimens of *Culex* (*Melanoconion*) sp. mosquitoes, were collected in São Paulo, Brazil, at the former São Paulo Zoological Park Foundation (currently Diretoria de Biodiversidade e Biotecnologia, DBB), at the Extra sampling point (23°38'48.2"S, 46°37'14.4"W), on October 9 and December 1, 2020 (Supplementary Fig. S1). Collections were carried out both at ground level (pool B38) and in the tree canopy (pool B39). Mosquitoes were captured using CDC light traps. Specimens were morphologically identified using dichotomous keys (Consoli and Lourenço-de-Oliveira, 1994; Forattini, 2002) at the Pasteur Institute in São Paulo, Brazil.

2.2. RNA extraction, sequencing, and library preparation

Total RNA was extracted from pooled mosquito samples using Maxwell 16 Viral Total Nucleic Acid Purification Kit (Promega, Inc., Madison, WI, USA) following the manufacturer's instructions. Libraries were constructed using the Nextera XT DNA library preparation kit (Illumina Inc., San Diego, CA, USA) and paired-end sequencing (2 × 150 bp) was performed on a MiSeq platform (Illumina Inc., San Diego, CA, USA) at the Central Laboratory of the Hospital das Clínicas, University of

São Paulo.

2.3. Quality control and preprocessing

Raw reads in FASTQ format were analyzed for quality using FastQC v.0.12.1 (Babraham Bioinformatics, 2025). Adapter sequences and low-quality regions were trimmed using Trimmomatic v.0.39 (Bolger et al., 2014), with distinct parameters for each library: Library 38: LEADING:3, TRAILING:3, SLIDINGWINDOW:4:20, HEADCROP:19, CROP:255. Library 39: LEADING:3, TRAILING:3, SLIDINGWINDOW:4:30, HEADCROP:19, CROP:235, MINLEN:210. Adapter file used: TruSeq2-PE.fa.

2.4. De novo assembly and viral detection

High-quality reads were assembled using rnaSPAdes (SPAdes RNA Viral) v.3.15.5 (Bushmanova et al., 2019), with default parameters suitable. The mean base coverage calculated specifically for the viral contigs was 37.038849 for pool B38 and 43.093060 for pool B39. The resulting contigs were visualized and curated using AliView v1.26 (Larsson, 2014).

Taxonomic annotation of contigs was performed using DIAMOND v.2.1.9 (Buchfink et al., 2015) against the NCBI non-redundant protein database (nr). Additional searches were conducted using BLASTN, BLASTX, BLASTP for similarity with nucleotides and proteins against the NCBI GenBank database (<https://www.ncbi.nlm.nih.gov/>).

2.5. Genome annotation and motif identification

The ORFs of the genomes were predicted using the online tool ORFfinder v.0.4.3 (<https://www.ncbi.nlm.nih.gov/orffinder/>). The motifs and domain were predicted using the Conserved Domains Tool v.0.4 (Lu et al., 2020) together with the Motif Finder (<https://www.genome.jp/tools/motif/>), respectively. To identify highly conserved motifs, the Palmscan algorithm was used (Babaian and Edgar, 2022).

2.6. Phylogenetic and distance analysis

The complete genome, capsid gene and RdRp sequences obtained from libraries 38 and 39 were compared with the closest sequences available in GenBank and ICTV reference sequences (ICTV, available at: <https://ictv.global/>) by multiple sequence alignment, using the MUSCLE algorithm (Edgar, 2004) implemented in the MEGA 12 software (Kumar et al., 2018). Phylogenetic trees (complete genome, capsid gene and RdRp) were synthesized using the maximum likelihood approach, with the statistical support of an ultra-fast bootstrap with 1000 interactions, using the best-fit models estimated with the software IQ-Tree (Minh et al., 2020). These were as follows: genome (GTR+F+R6), RdRp and Cap (VT+F+R7). The FigTree v.1.4.4 tool (<http://tree.bio.ed.ac.uk/software/figtree/>) was used to visualize the phylogeny and edit the visualization. Distances were calculated using MEGA 12 using the 'neighbor' method. Amino-acid identities were obtained using the SDT program v.1.2 (Muhire et al., 2025).

3. Results

3.1. Sample characterization

Two mosquito pools, labeled 38 and 39, collected from the São Paulo Zoo Foundation at the Ponto Extra site were analyzed in detail, focusing here on the identification and characterization of novel picorna-like viruses. These samples build on our previous viral metagenomic work aimed at characterizing mosquito-associated viral diversity in the region (Guimarães et al., 2025). The mosquito specimens analyzed in this study belong to the genus *Culex*, subgenus *Culex* (*Melanoconion*), and were sampled from two distinct strata: ground level and tree canopy

(Supplementary Table S1).

After processing the sequencing data, two contigs of nearly complete viruses were assembled, named *Culex (Melanoconion)* picorna-like virus CmV_B38 (pool 38; GenBank: PV576574) and *Culex (Melanoconion)* picorna-like virus CmV_B39 (pool 39; GenBank: PV576575). BLASTn and BLASTx non-redundant protein database searches showed higher identity, coverage, and e-value with Pyongtaek *Culex* virus (GenBank: MT568537) (Faizah et al., 2023) and Yongsan picorna-like virus 4 (GenBank: NC_040654) (Sanborn et al., 2019), respectively, not classified in the *Picornavirales* (Supplementary Tables S2 and S3).

3.2. Genome characteristics

The two genomes of *Culex (Melanoconion)* picorna-like viruses analyzed, CmV_B38 and CmV_B39, were identified as single-stranded RNA genomes with positive polarity. The CmV_B38 genome comprised 8582 nucleotides and exhibited a high A/U content (A: 30.04%; G: 22.04%; U: 29.09%; C: 17.03%). It contains a single ORF, spanning positions 69–8468 nt, which encodes a polyprotein of 2799 amino acids.

The CmV_B39 genome consisted of 8400 nucleotides and displayed a similar nucleotide composition (A: 30.05%; G: 22.05%; U: 29.07%; C: 17.03%). It also harbors a single ORF, located at positions 63–8400 nt, encoding a polyprotein of 2779 amino acids.

The analysis of the conserved domain with the aid of the NCBI CDD tools, MotifFinder and InterPro revealed that the genomes shared a genomic architecture typical of members of the order *Picornavirales* (Le Gall et al., 2008). The structural proteins are in the N-terminal region of the polyprotein, while the non-structural proteins are in the C-terminal region.

The non-structural proteins identified in the order include Hel, Pro, and RdRp. The helicase belongs to superfamily 3 (PF00910) and was detected in both sequences at positions 609–703 amino acids (aa), with an e-value of 6.1e-05.

The cysteine-like protease 3c (PF00548), was located at positions 1274–1307 aa (e-value: 0.012) contains a cysteine protease motif (GDCG) and a substrate binding motif (GIHVAG), both shared by CmV_B38 and CmV_B39.

The RdRp of RNA pol superfamily polymerase (PF00680), at positions 1411–1829 in CmV_B38 and CmV_B39, was predicted with four characteristic motifs of RdRp domain, including motif F (AFLKDELTSSTKIKAKTCRVIF), motif A (IAGDFKNFDQNM), motif B (SGCFFTTILNVLVH) and motif C (ILGDDHIY). In addition, two capsid domains related to Picornavirus (rhv_like; PF00073) were detected (Fig. 1).

3.3. Comparisons of *Culex (Melanoconion)* virus polyproteins

To better compare the amino-acid sequences of the CmV_B38 and CmV_B39 polyproteins, BLASTP searches were performed, resulting in the putative Yongsan picorna-like virus 4 polyprotein (GenBank: YP_009552779.1) as the “best hit” with a coverage value of 100%, identity of 73.31% (CmV_B38) and 71.14% (CmV_B39) between the sequences (Supplementary Table S4). It is worth mentioning that the Yongsan picorna-like virus 4 was also identified in mosquito samples,

however from the genus *Aedes*, species *Aedes vexans nipponii* (Sanborn et al., 2019).

3.4. Phylogenetic analysis

The maximum likelihood phylogenetic tree was inferred based on amino-acid sequences corresponding to the RdRp region, using cognate BLASTX sequences, reference sequences from the nine families of the order *Picornavirales* and sequences from PLVs and posaviruses identified in pig feces (posavirus), human feces (husavirus), panda (pansavirus), bat feces (basavirus), and rat feces-associated viruses (rasavirus). These viruses did not group with any of the nine currently recognized families of the order *Picornavirales* (Poch et al., 1989) (Fig. 2).

The *Culex (Melanoconion)* virus sequences (CmV_B38 and CmV_B39) were grouped within a clade (highlighted in yellow in Fig. 2) that was subdivided into two clades, A and B. Clade A was subdivided into two subclades, A1 and A2.

Subclade A1 was formed by Hubei picorna-like virus 9 (GenBank: YP_009337369) and Guiyang picorna-like virus 2 (GenBank: UHK03043). Subclade A2 contained the sequences of this study (CmV_B38 and CmV_B39) and three related viruses: Yongsan picorna-like virus 4 (GenBank: YP_009552779), Pyongtaek *Culex* picornavirus (GenBank: UGO57105 and BDV27094), and XiangYun picorna-like virus 5 (GenBank: UUG74230), with identities ranging from 46.6% to 99.79%. The average evolutionary distance between subclades A1 and A2 was 0.505, suggesting the possible presence of divergent lineages.

Clade B was formed exclusively by sequences of Basavirus 7 (GenBank: YP_009333263, APQ44487, APQ44490, APQ44496, APQ44493, and QVW10090) except for Hubei picorna-like virus 8 (GenBank: APG78038), which grouped in a separate branch. The distance between Clade A and Clade B was 0.577. Regarding the identity values between the members of Clade B, they ranged between 72.62% and 100% (Supplementary Fig. S2).

Phylogenetic trees including CmV_B38 and CmV_B39 were also inferred with the sequences showing the greatest homology in the RdRp tree, using genome and amino acids from the capsid region. The same topology was observed, as in the clade highlighted in yellow (Supplementary Figs. S3 and S4).

3.5. Comparison of conserved motifs A, B, and C of RdRp

For comparison purposes, the motifs of the RdRp region present in the CmV_B38 and CmV_B39 sequences were analyzed in relation to the Clade A and Clade B sequences using the PalmScan tool. The results indicated the presence of the three conserved RdRP motifs in the sequences, motifs A, B and C, with variations in the amino-acid residues (Fig. 3).

Motif A (12 aa) showed variation in the amino-acid sequence, with more frequent substitutions in the final positions, mainly involving residues Q, N and K. Motif B (14 aa) showed changes in position 8, with substitutions between I and P; in position 9, between I and L; in position 11, between I, V and C; and in position 12, between L and E. Motif C (8 aa) was the one that presented the least variations, including the GDD residues, characteristic of the RdRP active site, with variations only in position 1, where substitutions of amino acids I, V and L were observed.

The analyses demonstrate similarities and possible evolutionary divergences between the newly identified viruses and previously described taxa, indicating that CmV_B38 and CmV_B39 may represent new species within a group of viruses that do not yet have an established taxonomic classification.

4. Discussion

Building on our previous viral metagenomic analysis of adult female mosquitoes from the São Paulo Zoo Foundation, Brazil, which aimed to

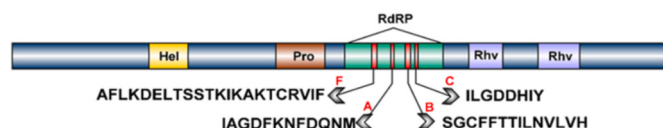


Fig. 1. Schematic organization of the CmV_B38 and CmV_B39 genomes. The polyprotein is represented by a dark blue box. The domains are highlighted by color (Helicase, yellow; Peptidase_C3, brown; RdRp domain, green). The RdRp domain includes four conserved motifs F, A, B, and C (red). Additionally, the capsid domains (Rhv) are also indicated in light blue.

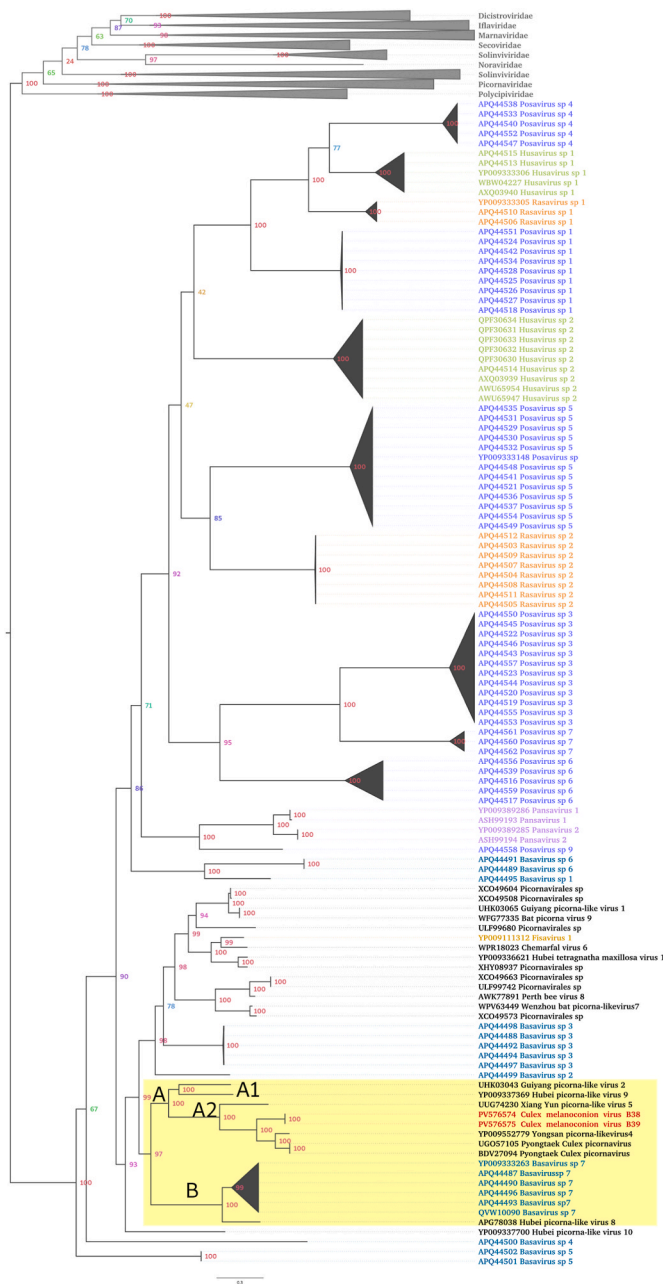


Fig. 2. Maximum likelihood tree of the RdRp region of the polyprotein. The clade containing sequences CmV_B38 and CmV_B39 is highlighted in yellow. This clade includes two subclades, A1 and A2, as indicated in the tree. Bootstrap support is indicated at the nodes. Sequences follow the format: GenBank ID and virus name. Posa-like viruses are highlighted in different colors: Posavirus (purple), Husavirus (green), Rasavirus (orange), Pansavirus (pink), Basavirus (blue), and Fisavirus (brown). The newly generated sequences CmV_B38 and CmV_B39 are highlighted in red. Clusters from the *Picornavirales* families were grouped to improve visualization.

expand knowledge of mosquito-associated viral diversity (Guimarães et al., 2025), this study provides a more detailed investigation of novel picorna-like viruses identified in *Culex (Melanoconion)* sp. mosquitoes. Polyprotein characterization of the identified viruses, CmV_B38 and CmV_B39, confirms features consistent with members of the order *Picornavirales*. Polyprotein analysis revealed the presence of five protein domains typical of viruses from this order, i.e. helicase, 3C-protease, RNA-dependent RNA polymerase (RdRp), and two rhv-like capsid domains (Le Gall et al., 2008).

In the RdRp region, four conserved motifs (F, A, B, and C) were identified, which are essential for viral replication and widely described in positive sense RNA viruses (Poch et al., 1989). However, despite this characteristic molecular signature, phylogenetic analysis demonstrated that the viruses do not group with any of the currently recognized families within the order *Picornavirales*.

These findings are particularly relevant in the context of arbovirus surveillance, since the identified viruses do not belong to classical arbovirus families (such as *Flaviviridae* or *Togaviridae*). Their detection in mosquitoes is of medical importance, as it suggests the role of these vectors as reservoirs of a viral diversity that is still poorly understood. Considering the current scenario of emergence of new arboviruses, such as Zika and chikungunya (Li et al., 2015; Yu and Cheng, 2022), the description of picorna-like viruses in mosquitoes expands the understanding of circulating viral diversity and highlights the importance of active surveillance (Shi et al., 2016; Cholleti et al., 2018).

The results of the amino-acid similarity analysis with virus sequences in GenBank showed that the “best hit” was Yongsan picorna-like virus 4 (GenBank: NC_040654), detected in *Aedes vexans nipponii* and *Culex pipiens* mosquitoes from the Yongsan military base in South Korea. Yongsan picorna-like virus 4 shares 75.98% amino-acid identity with the sequences described here in the RdRp region, in addition to presenting a similar genomic structure. Yongsan picorna-like virus 4 was the only one in the study to show a phylogenetic relationship with vertebrate viruses, although this relationship was distant compared to the close relationship of the invertebrate viruses analyzed (Sanborn et al., 2019). This highlights the evolutionary complexity of these viruses and their potential for cross-adaptation between hosts.

In addition to Yongsan picorna-like virus 4, several other picorna-like viruses have been identified in mosquitoes from the Republic of Korea, including Yongsan picorna-like virus 1 (GenBank: MH703050), Yongsan picorna-like virus 2 (GenBank: MH703060 and MH703052), and *Culex* picorna-like virus 1 (GenBank: MH703059), indicating a wide diversity of viruses of this order circulating in arthropods (Sanborn et al., 2019). These results, together with the growing number of picornavirus-like viruses detected in mosquitoes and other insects worldwide, support the hypothesis that Yongsan picorna-like virus 4 and the viruses identified in the present study (CmV_B38 and CmV_B39) represent mosquito viruses.

The identification of members of the order *Picornavirales* not yet classified in *Culex (Melanoconion)* broadens the scope that the virome composition can vary significantly among mosquito species of the family Culicidae, especially picorna-like viruses. It is important to highlight that the viruses whose sequences were described in Clade A showed genetic identities ranging from 46.6% to 99.79%. These viruses were identified in different arthropod hosts, such as Yongsan picorna-like virus 4 in *Aedes vexans nipponii* (Sanborn et al., 2019), Pyongtaek *Culex* picornavirus in *Culex* spp. (Faizah et al., 2023), XiangYun picorna-like virus 5 in *Anopheles sinensis*, in addition to Gulyang picorna-like virus 2 isolated from arachnids and Hubel picorna-like virus 9 detected in dragonflies (Shi et al., 2016). This indicates broad evolutionary diversity and possible adaptation to different arthropod hosts.

Analysis of RdRP motifs provided crucial evidence by demonstrating the conservation of motifs A, B, and C in Clade A members, which are highly conserved catalytic motifs among members of the order *Picornavirales* and essential for their enzymatic activity (Poch et al., 1989; Le Gall et al., 2008). The observed variation, particularly in motif B, indicates that these lineages may have undergone evolutionary adaptation, possibly associated with different mosquito host species or specific environmental selective pressures (Hulo et al., 2011; Jenkins et al., 2002; Koonin et al., 2020). These findings highlighted the geographical distribution and host diversity of this viral group, which was detected in countries such as South Korea, Japan and China (Shi et al., 2016; Sanborn et al., 2019; Faizah et al., 2023).

Evolutionary distance analysis between subclades A1 and A2 revealed a divergence of 50.5%, confirming that the separation is not the

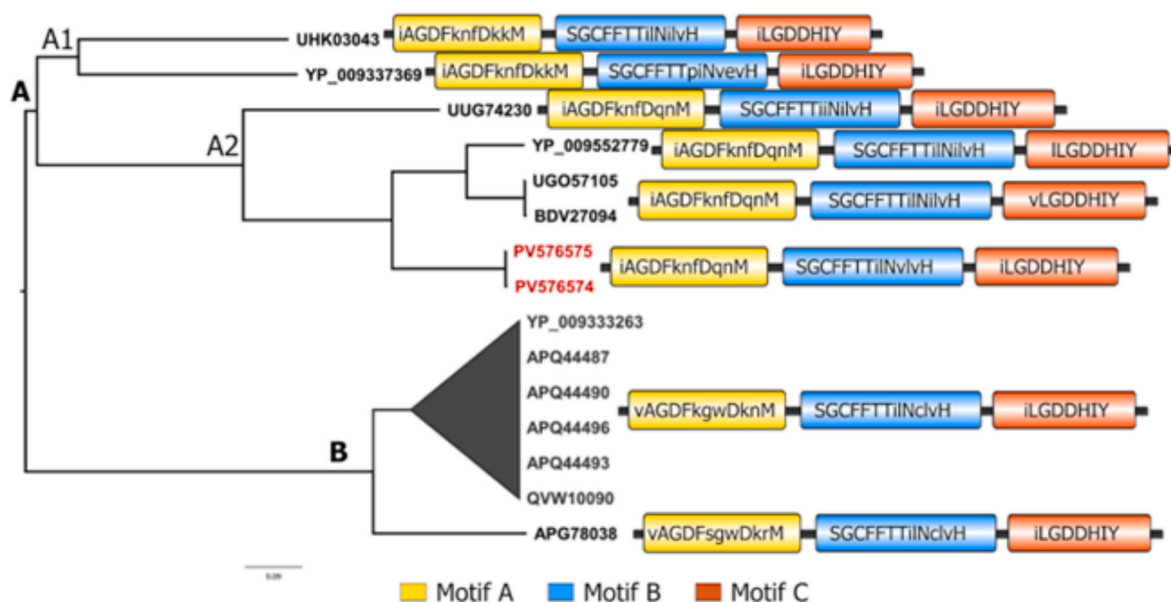


Fig. 3. Comparison of conserved motifs in RdRp sequences RNA-dependent RNA polymerases (RdRp) of CmV_B38 and CmV_B39 viruses with representatives of clades A and B. Clade A includes sequences from subclade A1 (YP_009552779, UGO57105, BDV27094, and UUG74230) and subclade A2 (YP_009337369 and UHK03043). Clade B consists of YP_009333263, APQ44487, APQ44490, APQ44496, APQ44493, QVW10090, and APG78038. The motif sequences are highlighted by color: motif A (yellow), motif B (blue), and motif C (red).

result of phylogenetic noise, a term that refers to random variations or methodological artefacts that can distort inferred evolutionary relationships (Townsend et al., 2012). The high statistical confidence confirms that this separation is not random, but reflects considerable evolutionary differentiation (Hillis and Bull, 1993).

These findings suggest the existence of genetically distinct lineages that may reflect adaptations to different hosts or environmental selective pressures, a phenomenon commonly observed in RNA viruses such as coronaviruses which exhibit high mutation rates (Montoya et al., 2021; Warger and Gaudieri, 2022).

The evolutionary distance between Clades A and B was 0.577, and it is important to emphasize that most Clade B viruses (basavirus) were identified in bats. Bats, recognized as natural reservoirs for several viruses, have unique ecological and physiological characteristics that may influence viral evolution. The significant genetic distance observed between Clades A and B, coupled with their distribution in distinct hosts, suggests that these groups may have followed distinct evolutionary trajectories, possibly related to adaptations to different ecological niches or specific transmission mechanisms (Denton et al., 2013; Brook and Dobson, 2015; Plowright et al., 2015).

Oude Munnink et al. (2017) investigated picorna-like virus genomes and demonstrated that they belonged to distinct groups from genomes already classified in the order *Picornavirales*, with different levels of nucleotide identity (60–90%), suggesting a provisional identification until new data allow a more precise classification. These findings reinforce the possibility of the existence of a wide diversity of picornavirus-like viruses in arthropods, as also observed in the present study.

5. Conclusion

The viral sequences detected in *Culex (Melanoconion)* mosquitoes expand our knowledge of the diversity of previously unclassified viruses within the order *Picornavirales* in mosquitoes. The conservation of RdRp motifs and the observed divergence between Clade A subgroups indicate genetic variability, possibly related to different hosts and environments. These findings highlight the importance of characterizing viruses in mosquitoes for understanding viral evolution and dispersal.

Ethical approval

Not applicable.

CRediT authorship contribution statement

Lilian de Oliveira Guimarães: Conceptualization, Data curation, Formal analysis, Methodology, Writing – review & editing. Ariel Rocha de Almeida: Formal analysis, Writing – review & editing. Endrya do Socorro Foro Ramos: Formal analysis, Writing – review & editing. Juliana Telles-de-Deus: Formal analysis, Writing – review & editing. Vanessa Christe Helfstein: Formal analysis, Writing – review & editing. Vanessa dos Santos Moraes: Formal analysis, Writing – review & editing. Jesus Maia dos Santos: Formal analysis, Writing – review & editing. Ramendra Pati Pandey: Writing – review & editing. Vera Lucia Fonseca de Camargo-Neves: Data curation, Funding acquisition, Resources, Writing – review & editing. Antonio Charlys da Costa: Conceptualization, Formal analysis, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing. Karin Kirchgatter: Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing. Élcio Leal: Project administration, Supervision, Writing – original draft, Writing – review & editing.

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Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.crvbd.2025.100333>.

Data availability

The data supporting the conclusions of this article are included within the article and its supplementary files. The newly generated sequences were submitted to the GenBank database under the accession numbers PV576574 and PV576575.

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